Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/30/2026
PI Name:	Mortreux, Marie Ph.D.		
Project Title:	Circadian Rhythm Disruption and Gravitational Disturbance in a Lunar Mission Analog: Consequences for Muscle Function During and After the Mission		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
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No. of PhD Candidates:		**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2025 per the PI (Ed., 6/2/25). NOTE: End date changed to 05/31/2025 per F. Hernandez/ARC (Ed., 6/21/2024).		
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Task Description:	Mechanical loading is the primary stimulus required for the maintenance and health of the musculoskeletal system. Exposure to microgravity or reduced gravity results in rapid bone loss and muscle atrophy, especially in the lower limbs. In recent years, the clock system in the skeletal muscle has been recognized to play a critical role in key aspects of muscle physiology, ranging from structural maintenance to functional regulation. On Earth, perturbations of the circadian rhythm affect muscle function and are associated with the development of sarcopenia. Reciprocally, studies have demonstrated that muscle activity can directly modulate the expression of muscle clock genes in a time-dependent fashion, and recent data suggest that the circadian clock could influence skeletal muscle adaptation in response to exercise training. Since Artemis astronauts will experience alterations in muscle condition and circadian rhythm simultaneously, it is critical to develop ground-based studies that will closely mimic this situation. While these stressors will occur temporarily, their effects may linger, and negatively influence muscle recovery after return from the mission. In this ground-based Early Career Investigator (ECI) proposal, I hypothesize that light cycle disturbances induced by a model of Chronic Jet Lag (CJL) will severely impact the peripheral muscle clock and, combined with altered gravity, will lead to additive negative effects during the disuse period. Moreover, I hypothesize that circadian disruption will have long-lasting effects that will significantly impair muscle recovery during the reloading period. SA1: To characterize the impact of circadian disturbance and partial gravity in a lunar analog in male and female rats. 60 adult outbred rats (30/sex) will undergo 7 days of exposure to partial weight-bearing at 20% of normal loading (PWB20) to simulate lunar gravity. 1 group will be exposed to a regular light cycle (LD), 1 group will be kept in constant light to induce circadian free-running (LL), and the experimental group will be exposed to CJL using a 6h phase advance every other day. SA2: To determine and compare the long-term effects of circadian disturbance and altered gravity during muscle recovery in males and females. 20 animals (10/sex) will be exposed to the same PWB20+CJL paradigm as in Aim 1. Animals will be allowed to recover for 7 days at normal loading (1g) and under a standard light cycle (LD). For this work, I plan on first assessing muscle health and function through regular testing including grip strength, fatigue resistance, force production, and muscle quality. I will also monitor circadian parameters and rest/activity over several 24h periods. Finally, I will perform biomolecular assays in the suprachiasmatic nucleus and soleus muscle, targeting specific genes involved in circadian rhythm entrainment, muscle function, and inflammation. This promising pilot study will help assess the additive effects of circadian disturbance in animals exposed to partial gravity and will help determine the existence of sex-based differences in response to spaceflight stressors. Moreover, I will determine sex-based differences during muscle recovery following disuse, and the long-term impact of circadian disturbance on muscle health and function. This study, which includes phenotypical and molecular outcomes, will assess the contribution of muscle clock genes and pathways to muscle health. Finally, targeted molecular assessment will help me use my diverse background (metabolism, neuromuscular, circadian) to explore other systems of interest that may be linked to muscle health and function (e.g., metabolic signaling pathways), and help characterize the key players that could be targeted with pharmacological approaches to provide new and effective countermeasures in future studies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research explores the combined effect of partial gravity and circadian disturbances on muscle health and if it impacts recovery. This work may help provide insights regarding the health and performance of the Artemis astronauts who will experience reduced gravity and light disturbances during lunar missions, and may lead to the development of new strategies and countermeasures for astronauts and people on Earth performing shift-work or experiencing chronic jet lag.
Task Progress:	This project aims to answer two main questions: 1) How will the lunar stressors (gravity and light disturbances) impact muscle health and function in males and females?, and 2) How will these stressors impact recovery after the mission? We are reporting some results here, but findings are not yet published and this report will be updated with more details following publication. To answer these questions, male and female adult rats were exposed to simulated lunar gravity (20% of Earth gravity) for one week, and subjected to different light conditions (standard cycle, constant light, or chronic jet lag). Animals exposed to simulated lunar and chronic jet lag were also allowed to recover for one week in normal conditions (100% of Earth gravity and standard light cycle). We monitored muscle strength, function, electrophysiological, and contractile properties throughout the study. We also assessed fine motor skills, and obtained muscle mass. Muscle strength was reduced following simulated lunar gravity, more so in males than females, confirming the results of previous studies. The maximal force produced by the calf muscles was significantly impacted by the light cycles, and contractility studies highlighted changes in myofiber type composition impacting muscle fatigability. Changes in muscle electrophysiological properties are well characterized in males, however our results highlight that females experienced larger variations in standard light conditions. Surprisingly, soleus wet mass did not always correlate with muscle strength, and males displayed the largest variations. In females, fine motor skills were compromised in animals exposed to chronic jet lag, while males were not. Several circadian parameters were followed, including temperature, grip strength, and blood glucose. While animals in standard conditions experience regular cycles throughout a 24h period, we noticed significant shifts in response to lighting conditions. When animals were allowed to recover in normal conditions, muscle strength was recovered comparable to baseline levels. However, circadian parameters followed over a 24h period remained impacted in both males and females, suggesting recovery was incomplete due to the light disturbances. Contractility studies highlighted lingering impacts of lunar stressors on muscle performance, and muscle mass was not fully recovered after one week.
Bibliography Type:	Description: (Last Updated: 05/04/2026)

Abstracts for Journals and Proceedings	Dworacek S, Ayi-Bonte D, Dunlap K, Madden J, Nigro E, Mortreux M. "Circadian rhythm disruption and gravitational disturbance in a lunar mission analog, a study in male and female rats." 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. Abstracts. 40th Annual Meeting of the American Society for Gravitational and Space Research, San Juan, Puerto Rico, December 3-7, 2024. , Dec-2024